

EMBEDDED SYSTEMS PROJECT PROPOSAL

Smart Street Light System

ATmega328P | Microchip Studio 7 | Proteus 8 | Embedded C
Automatic Adaptive Lighting with PIR Motion Detection & PWM Brightness Control

1. Project Title

Smart Street Light System Using AVR Microcontroller (ATmega328P)

2. Team Details

Field	Details
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3. Abstract

This project proposes the design and simulation of an **Automatic Smart Street Light System** using the ATmega328P AVR microcontroller, programmed in Embedded C via Microchip Studio, and simulated using Proteus 8 Professional. The system automates street light operation based on ambient light levels detected by an LDR (Light Dependent Resistor) and motion/vehicle presence detected by a PIR sensor. During daytime, lights remain OFF; at night, lights switch to low-brightness mode. When motion is detected, the relevant light switches to full brightness and gradually dims after the motion clears. **Unlike basic solar street lights that only perform simple ON/OFF switching**, this system introduces adaptive intelligence — responding to real-time conditions, reducing energy consumption by up to 60–70% while maintaining safety. This solution is cost-effective, scalable, and directly applicable to smart city infrastructure.

4. Problem Statement

Traditional street lighting systems suffer from several inefficiencies:

- Lights remain ON throughout the night at full brightness, even in low-traffic periods, wasting significant electrical energy.
- Manual ON/OFF switching requires human intervention, making it unreliable and prone to delays.

- No adaptive response to real-world conditions such as ambient light changes or pedestrian/vehicle presence.
- High energy bills and carbon footprint due to lack of intelligent control mechanisms.

Current solutions such as simple timer-based systems do not account for dynamic environmental changes. A sensor-driven, microcontroller-based embedded system is required to address these issues effectively.

Why existing solar lights are NOT sufficient:

Solar street lights solve the energy SOURCE problem (using solar panels) but NOT the energy WASTE problem. They still burn at full brightness all night regardless of whether anyone is present. This project solves BOTH — it can work with any power source AND intelligently adjusts brightness based on actual real-time need.

5. Objectives

The key objectives of this project are:

- **Design** an embedded system using the ATmega328P AVR microcontroller that automates street light control.
- **Implement** ambient light detection using an LDR to switch lights ON/OFF automatically based on day/night conditions.
- **Integrate** a PIR motion sensor to enable adaptive brightness — full brightness on motion detection, dimming after a timeout.
- **Develop** firmware in Embedded C using Microchip Studio, utilizing timers, interrupts, ADC, and GPIO.
- **Simulate** the complete circuit and firmware behavior in Proteus 8, verifying correct functionality before hardware deployment.

6. System Overview

The Smart Street Light System works as follows:

Stage	Description
INPUT	LDR sensor reads ambient light level (ADC input). PIR sensor detects motion/heat signature (digital interrupt).
PROCESS	ATmega328P reads ADC values from LDR. Compares against threshold to determine day/night. On motion, triggers PWM brightness boost via Timer1. After configurable timeout (30 seconds), returns to dim mode.
OUTPUT	LED street lights controlled via PWM (Timer1, Fast PWM mode). LCD 16x2 (optional) displays current mode and status. Buzzer (optional) for fault alert.

7. System Design

7.1 Block Diagram

Conceptual block diagram description:

- Power Supply (5V DC) → Voltage Regulator → ATmega328P
- LDR + 10kΩ resistor voltage divider → ADC0 (PC0) of ATmega328P
- PIR Sensor Output → INT0 (PD2) External Interrupt pin
- ATmega328P PWM Output (OC1A / PB1) → MOSFET Gate → LED Street Light
- Optional: LCD data pins → PORTD, LCD control → PORTB

7.2 Components List

Component	Model / Value	Qty	Function
Microcontroller	ATmega328P (DIP-28)	1	Main control unit
LDR Sensor	GL5528	1	Ambient light detection
PIR Motion Sensor	HC-SR501	1	Motion/presence detection
LED (Street Light)	High-brightness white LED / 5mm	3–5	Light output
N-Channel MOSFET	2N7000 / IRF540	1	PWM-controlled LED driver
Resistors	10kΩ, 330Ω, 1kΩ	Multiple	Voltage dividers, current limiting
Crystal Oscillator	16 MHz + 2×22pF capacitors	1	External clock for ATmega
LCD Display	16×2 (HD44780)	1 (opt.)	Status display
Buzzer	5V Active Buzzer	1 (opt.)	Fault alert output
Power Supply	5V DC Regulated	1	System power

8. Software Approach

8.1 Development Environment

- **IDE:** Microchip Studio 7 (formerly Atmel Studio)
- **Language:** Embedded C (C99 standard with AVR-GCC toolchain)
- **Simulation:** Proteus 8 Professional (ISIS schematic + VSM simulation)
- **Programmer:** AVR Dragon / USBasp (for physical hardware, if required)

8.2 Key Firmware Concepts

Concept	Application in Project
ADC (Analog to Digital Conversion)	Read LDR voltage on ADC0; compare against threshold to detect day/night.
External Interrupt (INT0)	Detect rising edge from PIR sensor output; trigger ISR to boost LED brightness.
Timer1 – Fast PWM	Generate PWM signal on OC1A (PB1) to control LED brightness (0–255 duty cycle).

Timer0 – CTC Mode	Generate delay/timeout counter to dim lights after motion clears (30s timeout).
GPIO (Digital I/O)	Control LCD, buzzer, and indicator LEDs via PORTB/PORTD.
Sleep Modes	Put MCU in Idle Sleep between ADC readings to reduce microcontroller power draw.

8.3 Firmware Logic (Pseudocode)

```
// Initialize ADC, Timer0, Timer1 (PWM), INT0 (PIR interrupt), LCD
Set brightness = LOW_DIM (OCR1A = 50)

// Main Loop:
ldr_val = ADC_Read(ADC0);
IF ldr_val > DAY_THRESHOLD: → set_brightness(0) // Daytime: LEDs OFF
ELSE (night):
    IF motion_flag == 1: → set_brightness(255) // Full brightness
        timeout_cnt++ → if >= 30s: dim again
    ELSE: → set_brightness(60) // Dim mode
Update LCD: DAY / NIGHT DIM / NIGHT MOTION
```

9. Methodology

Phase	Activity	Tool
1. Design	Schematic design, component selection, pin mapping	Proteus 8 ISIS
2. Firmware Dev	Write Embedded C code for ADC, PWM, interrupts	Microchip Studio 7
3. Compilation	Compile with AVR-GCC, generate .hex file	Microchip Studio 7
4. Simulation	Load .hex into Proteus virtual ATmega328P, run simulation	Proteus 8 VSM
5. Testing	Vary LDR light level, trigger PIR — verify LED behavior	Proteus 8
6. Debugging	Use Proteus logic analyzer / oscilloscope to debug PWM	Proteus 8 Tools
7. Documentation	Write report, capture screenshots, finalize block diagram	MS Word / Draw.io

10. Expected Outcome

- A fully functional Proteus 8 simulation of the Smart Street Light System, demonstrating:
 - Automatic light ON/OFF based on LDR ambient light level
 - Adaptive brightness control using PWM when motion is detected by PIR sensor
 - Return to dim mode after configurable timeout period
 - Optional LCD displaying current system mode in real time
- **Clean, well-commented Embedded C firmware** developed in Microchip Studio with proper use of AVR peripherals (ADC, Timer0, Timer1, INT0).

- A documented block diagram and circuit schematic ready for hardware prototyping on ATmega328P.
- Demonstrated reduction in simulated energy consumption compared to always-on lighting.

11. Applications

- **Smart City Infrastructure:** Adaptive street lighting in residential and commercial areas, reducing electricity costs by 40–60%.
- **Highway and Tunnel Lighting:** Automated brightness adjustment based on traffic density and ambient conditions.
- **Parking Lots and Campuses:** Motion-triggered lighting for security and energy efficiency in low-traffic night-time zones.
- **Industrial Zones:** Automated perimeter lighting that activates only on personnel or vehicle movement.
- **Future IoT Extension:** UART/GSM module can be added to ATmega328P for remote monitoring and fault reporting — upgrade path toward full smart city integration.

12. Innovation & Differentiation from Existing Solutions

12.1 How This System Differs from Commercial Solar Street Lights

Commercially available solar street lights address the energy source problem by replacing grid power with solar panels. However, they do not fundamentally solve the problem of **intelligent, adaptive light control**. Most budget solar street lights operate in a simple binary ON/OFF fashion triggered by ambient darkness, with no awareness of motion, traffic, or real-time environmental context. This project targets a different and complementary dimension of innovation: **the intelligence layer** that governs how and when light is delivered, regardless of the power source.

12.2 Key Points of Innovation

★ 1. PWM-Based Adaptive Brightness Control

Unlike conventional solar lights that are fully ON or fully OFF, this system uses Timer1 in Fast PWM mode to deliver three distinct brightness levels: OFF (daytime), DIM (night, no motion), and FULL BRIGHT (night, motion detected). This graduated response reduces energy consumption during low-traffic hours while ensuring maximum visibility when needed. Standard solar lights do not implement variable duty-cycle PWM control at the firmware level.

★ 2. Dual-Sensor Fusion (LDR + PIR)

The system combines two independent sensing modalities: ambient light (LDR via ADC) and motion/heat signature (PIR via external interrupt INT0). The decision logic uses both inputs simultaneously to determine the correct output state. This sensor fusion approach is absent in most off-the-shelf solar solutions, which rely solely on a photoresistor for switching.

★ 3. Configurable Timeout with Timer0 (CTC Mode)

After motion is no longer detected, the system does not abruptly cut power. Instead, Timer0 in CTC mode implements a configurable hold-on timeout (e.g., 30 seconds) before returning to dim mode. This prevents rapid flickering in low-traffic scenarios and improves user safety. This behavior requires firmware-level timer management not present in simple solar light controllers.

★ 4. MCU Sleep Mode for Microcontroller Power Efficiency

The ATmega328P is placed into Idle Sleep mode between ADC sampling cycles. This reduces the microcontroller's own power draw significantly during idle periods — a hardware-level optimization invisible in commercial products but a meaningful engineering design decision in resource-constrained embedded systems for smart city deployment.

★ 5. Fault Alerting via Buzzer Output

An optional buzzer output is included to signal system faults such as sensor malfunction or unexpected state conditions. This fault-detection layer represents a reliability feature not present in standard solar lights, making the system more suitable for unattended smart infrastructure deployment.

★ 6. Complementary with Solar Power Source

This system does not compete with solar-powered street lights — it enhances them. The intelligent control firmware can be directly integrated into a solar-powered street light to produce a system that is both energy-source independent AND intelligently adaptive. The innovation presented here is therefore orthogonal to, and stackable with, existing solar technology.

12.3 Comparative Summary

Feature	Basic Solar Light	Timer-Based Light	This Project (ATmega328P)
Ambient Light Detection	Yes	No	Yes (ADC + LDR)
Motion Detection	No	No	Yes (PIR + INT0)
Adaptive PWM Brightness	No	No	Yes (Timer1 Fast PWM)
Configurable Timeout	No	Partial	Yes (Timer0 CTC)
MCU Sleep / Power Saving	No	No	Yes (Idle Sleep Mode)
Fault Alert Output	No	No	Yes (Buzzer)
Solar-Compatible	Yes	No	Yes (5V compatible)

Summary:

While solar street lights solve the energy source problem, this project solves the intelligent control problem. The two innovations are complementary, and this system can be deployed alongside or integrated into solar-powered infrastructure to produce a complete smart street lighting solution.

13. References

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